Ziyi Sun

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College of Engineering, Ocean University of China, Qingdao 266100, China

Education

- **Ocean University of China (985 / Double First-Class)** *Sep. 2023 – Jun. 2027 (Expected)*
  B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
  - GPA: **3.7/4.0** Rank: **1/71** Average: 89.2
  - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

Research Experience

- **Roboparty Lab** *Jul. 2026 – Oct. 2026*
  Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
  - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

Projects

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** *Dec. 2025 – Dec. 2026*
  Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
  - Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.
  - Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** *Dec. 2024 – Nov. 2025*
  Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection Provincial Innovation Program
  - Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.
  - Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.
- **Modular Adaptive Deep-Sea Resource Collection and Locomotion Platform** *Dec. 2024 – Nov. 2025*
  Role: Core Member | Tools: Fluent, mechanical design Research Project
  - Designed a modular adaptive deep-sea mining vehicle for complex seabed terrain, including Fluent-based plume analysis during locomotion/mining and plume-suppression structures.
  - Combined an active four-track chassis with a compliant collection structure to improve geological adaptability on the seabed.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** *Dec. 2024 – Nov. 2025*
  Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National Innovation Program
  - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

Publications & Patents

C=Conference, S=In Submission, P=Patent, SW=Software

[C.1] **A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot**.
Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.
IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist**.
- Proposed a two-stage framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization.
- Achieved 70%–90% success over 60 real water-tank trials.

[C.2] **Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator**.
Dingyang Yu, Yuanrong Zhong, Qi Chen, Siming Ke, Qianbei Luo, ***Ziyi Sun***, Xunqi Zhao.
IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.

◦ Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

[S.1] CHOREO: Every Humanoid Skill as a Traject.
Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.
Under review at *AAAI 2027* (CCF-A).

◦ Unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills.

[S.2] A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.
Ziyi Sun, Changhong He, et al.
Under review at *Journal of Marine Science and Engineering (JMSE)*.

◦ Developed a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[P.1] A High-Stability PTO System for Duck-Type Wave Energy Converters.
Zhi Han, ***Ziyi Sun***, Haoran Chen, Hailong Zhao, Wenchi Wei, Xueyao Wang, Hongda Shi.
Invention Patent (First Student Inventor).

[SW.1] Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.
Haoyuan Cheng, Chuanyun Su, Zhichao Yan, Shuning Wang, ***Ziyi Sun***.
Software Copyright (No. 2024SR0140235).

Skills

- **Robot Platform:** Unitree G1
- **Programming & ML:** Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:** SolidWorks, AutoCAD
- **Languages:** Chinese (Native); English (CET-4: 591, CET-6: 482)

Honors and Awards

- **Shandong Provincial Government Scholarship** — *2025*
- **First-Class Comprehensive Scholarship, Ocean University of China** — *2024 & 2025*
- **Outstanding Student / Outstanding Student Cadre, Ocean University of China** — *2024 & 2025*
- **National First Prize, National 3D Digital Innovation Design Competition** — *2025 & 2026*
- **National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)** — *2025*
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** — *2025*
- **National Third Prize, 18th National College Student Energy Conservation Competition** — *2024*
- **National Gold Award, 3rd "Chuangyi Cup" National Undergraduate Extracurricular Academic Contest** — *2024*
- **National Second Prize, 14th Marine Vehicle Design and Production Competition** — *2025*
- **National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition** — *2026*
- **Science and Technology Innovation Pioneer Team, College of Engineering, OUC** — *2024*

Leadership Experience

- **Ocean University of China** — *2023 – Present*
  Class Representative / Class Monitor — Qingdao, China
- **College of Engineering Debate Team, OUC** — *2024 – 2025*
  Team Leader — Qingdao, China
- **Student Association for the Study of Socialism with Chinese Characteristics, OUC** — *2024 – 2025*
  Director of Event Organization — Qingdao, China

Volunteer Experience

- **Volunteer,** Qingdao Marathon — *2023*
- **Member,** National College Student Volunteer Outreach Team — *2024*
- **Outstanding Individual in Summer Social Practice,** OUC — *2024*